

Section 03



Industrial Security (Bachelor)

Section 03 – Industrial Network Architectures

Industrial Network Architectures

Lecture Notes

Department of Computer Science

1 Purdue Reference Model

2 Topologies

3 Zones and Conduits

By the end of this section you will be able to

- Apply the Purdue Enterprise Reference Architecture.
- Explain the role of the Industrial DMZ.
- Compare common plant-floor topologies and their trade-offs.
- Apply the IEC 62443 concept of zones and conduits.
- Identify when to use a data diode or unidirectional gateway.

1

Purdue Reference Model

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Source: T.J. Williams, "The Purdue Enterprise Reference Architecture", *Computers in Industry*, 1994 | ISA-95 / IEC 62264.



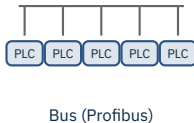
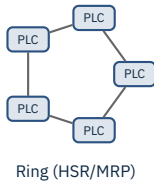
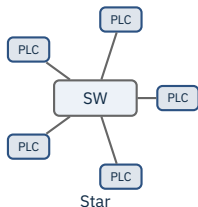
Iron rule

No direct path between IT and OT. Data *lands* in the DMZ; consumers *pull* from there.

2

Topologies

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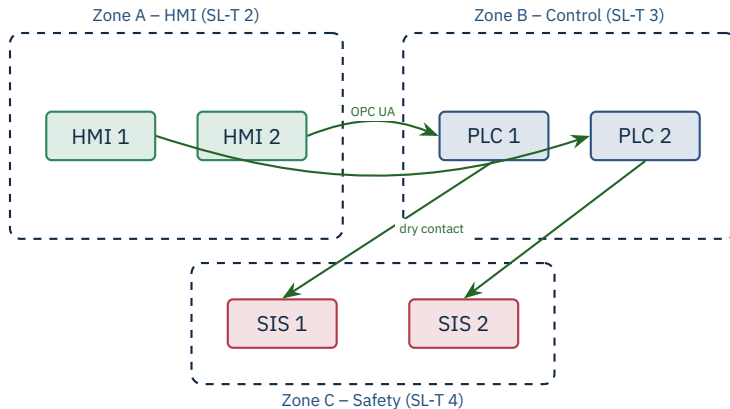
Driver

Topology choice in OT is driven by **determinism** and **availability**, not security.

3

Zones and Conduits

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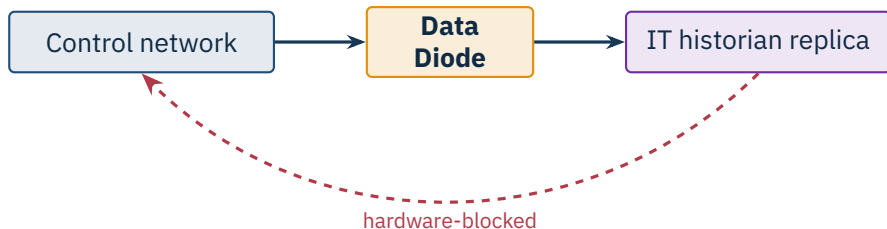
Zones group assets with similar security needs; conduits are the controlled channels between them.

Modbus function-code allowlist
read coils ✓ write multiple × (after-hours)



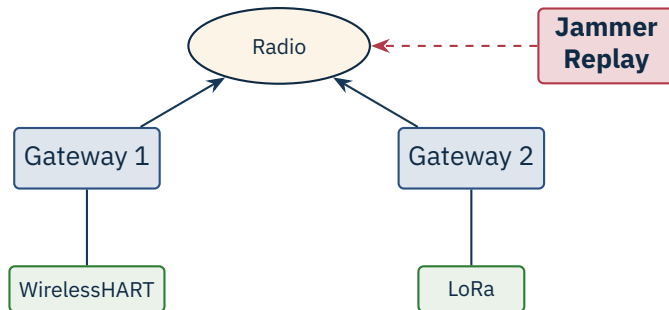
Industrial firewalls do stateful inspection *and* ICS protocol parsing. Often deployed transparently.

Source: Examples: Cisco ISA-3000, Fortinet Rugged, Belden Hirschmann Tofino, Palo Alto IoT Security.

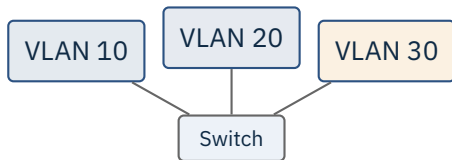


Physical fibre layout enforces the direction. Useful for safety-critical data egress.

Source: Vendors: Owl Cyber Defense, Waterfall Security Solutions, Fox-IT DataDiode | certified to IEC 62443-3-3 SR 5.1.

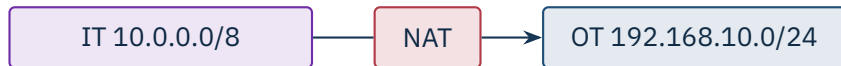


Convenience vs. attack surface. Never put safety-critical loops over uncontrolled wireless.



Caution

VLAN hopping is a real attack. A misconfigured trunk port (DTP enabled, native VLAN guessed) lets an attacker hop between VLANs. Use VLANs for organisation; require a routed firewall for security boundaries.



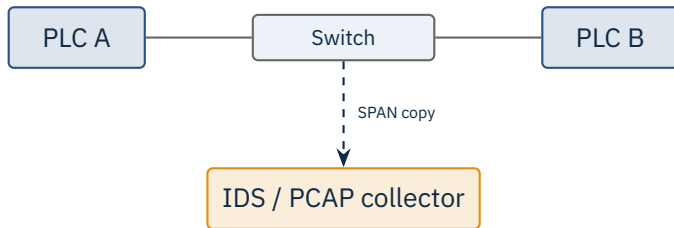
NAT hides, it does not protect

NAT breaks end-to-end addressability – useful for legacy duplicate networks, harmful for incident analysis. Prefer routed, segmented design with explicit firewall rules.



Why it matters for security

Without synchronised clocks, alarm logs and PCAPs are useless in forensics. PTP (IEEE 1588) gives sub-microsecond accuracy on the plant floor; NTP is enough for L3+.



SPAN vs hardware TAP

SPAN can drop packets under load and lets the switch see your collector. A hardware TAP is read-only by construction – preferred for high-fidelity capture.

Tech	Latency	Range	Typical use
Wi-Fi 6 / 6E	1–10 ms	10–100 m	Mobile HMI, video
WirelessHART	250 ms	100 m	Process sensors
ISA-100.11a	250 ms	100 m	Process sensors
LoRaWAN	seconds	km	Building automation
NB-IoT / LTE-M	seconds	km	Remote telemetry
Private 5G	1 ms	100–500 m	High-density factory

★ Key Takeaway: What to remember from this section

- The Purdue model groups assets into layers; firewalls separate the layers.
- The Industrial DMZ brokers all traffic between corporate IT and the plant.
- Zones and conduits give a structured language for IEC 62443 design.
- Hardware mediation (diodes, gateways) protects the most sensitive flows.

- Williams, T.J. *The Purdue Enterprise Reference Architecture*, Computers in Industry, 1994.
- ANSI/ISA-95 (IEC 62264). *Enterprise-Control System Integration*.
- IEC 62443-3-2:2020. *Security risk assessment for system design (zones & conduits)*.
- IEC 62439-3:2021. *High Availability Seamless Redundancy (HSR) and Parallel Redundancy Protocol (PRP)*.
- IEEE 1588-2019. *Precision Time Protocol (PTP)*.
- IEC 62591:2016. *WirelessHART*.
- IEC 62734:2014. *ISA-100.11a wireless system for industrial automation*.
- Idaho National Lab. *Recommended Practice: Improving Industrial Control System Cybersecurity with Defense-in-Depth Strategies*, 2016.
- CISA. *Securing Network Infrastructure Devices*, 2018.
- Stouffer, K. et al. *NIST SP 800-82 Rev. 3 – Guide to OT Security*, Sep 2023.

End of Section 03



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Questions and discussion